samples that are taken, the smoother-looking the result will be, but will require extra time spent in rendering.

Blender's Material UI

In order to create great-looking material in Blender, it would be useful if you shift to the material layout. Access this interface using the drop-down menu from the top centre part of Blender. Figure 1 shows the Material UI.

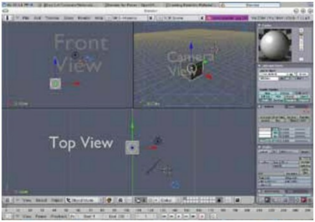

Figure 1: Material UI

A brief look at the main shaders

The Blinn-Phong reflection model for plastics: The Phong reflection model (also called Phong illumination or Phong lighting) is an empirical model of the local illumination of the points on a surface. The Blinn-Phong shading model (also called the Blinn–Phong reflection model or the modified Phong reflection model) is a modification to the Phong reflection model by Jim Blinn.

Lambertian reflectance on non-reflective surfaces: A surface is said to exhibit Lambertian reflectance when light falling on it is scattered such that the apparent brightness of the surface to an observer is the same regardless of the observer's angle of view. To phrase it in more technical terms, the surface luminance is isotropic. For example, unfinished wood exhibits roughly Lambertian reflectance, but wood finished with a glossy coat of polyurethane does not, since specular highlights may appear at different locations on the surface. Not all rough surfaces are perfect Lambertian reflectors, but this is often a good approximation when the characteristics of the surface are unknown. Lambertian reflectance is named after Johann Heinrich Lambert.

The Cook-Torrance shader: The Cook-Torrance shader was an adaptation of a previous method proposed by the applied physicists Torrance and Sparrow. This method is often used to simulate the reflective properties of metals and plastics.

The Oren-Nayar model for rough surfaces: The Oren-Nayar reflectance model, developed by Michael Oren and Shree K Nayar, is a model for diffusing reflection from rough surfaces. It has been shown to accurately predict the appearance of a wide range of natural surfaces, such as concrete, plaster, sand, etc.

Creating materials in Blender

A brief look at the Material Node Editor

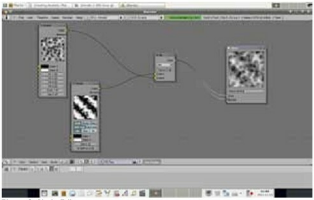

Figure 2: Node Editor

While applying pre-made textures may be a comparatively easy task, it is not so when creating materials. Creating realistic-looking material becomes a real challenge for 3D artists, who have to bring in realism into their scenes and characters. There are various ways of creating materials in Blender. Once you study the material interface, it becomes fairly easy to handle the creation of materials. While simple materials can be created by clicking a few buttons, complex materials require the use of material nodes, where connections and interconnections are made to simulate very complex-looking materials. Figure 2 shows the Node Editor while working with Material Nodes. You can access the Node Editor using the keys Shift+F3. The figure shows two in-built Blender textures, Voronoi and Marble, which have been combined using the material nodes to produce a different effect. Thousands of such permutations and combinations could be made in Blender to create complex materials, which are rather difficult to create in any other 3D application.

Simulate simple materials in the Blender Material UI

Creating materials in Blender can be a daunting task for a novice. But if you follow a little science and understand the system by which Blender works, you will be able to create wonderful materials in no time. Let's quickly go through a brief exercise to understand how to create simple yet realistic materials within Blender, without having to use the Node Editor. The materials we will create are glass, brass, wax and iron.

Fire up Blender and shift to the Material UI. You will observe three view-ports: the *Front*, *Top* and the *Camera* views. Toward the right is an array of buttons related to materials creation, normally accessed using F5 on the keyboard.

Creating glass: Once you are comfortable with the Material UI, get started with the first material, glass. Before modifying the button settings to simulate glass, consider its features. First, it is transparent; second, it is reflective. More important still is the refractive index of glass. We know it could